# Adaptive evaluation of the subexponential SAT mode graph

Smaller certified exponential bounds, a review of twenty packets, and a conserved degree budget for further research

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**Abstract**

We review the fifteen integrated refinement packets and all five incoming SAT archives at ProveIt commit `fd0edd17e6a2`. We obtain two smaller explicit upper bounds within their shared exact tensor-contraction framework. Adaptive four-block separator conditioning gives

$$T \leq (s+\mathcal{L})^{O(1)} 2^{(50.30468921410+o(1))s/\sqrt{\log_2 s}}$$

in polynomial space. A covariance bound controlling all three block constructions and the final matrix contraction simultaneously gives time and space coefficient 31.25239731266 on the same scale. The strongest previously displayed coefficients are 55.70947831891 in polynomial space and 35.35458805840 with subexponential space. We also compare against a stronger 55.40466590661 polynomial-space composition of incoming results.

These are reductions of the leading coefficient in an exponential upper bound. We do not obtain a smaller order than $n \log\log n/\sqrt{\log n}$ for general-density SAT, or $n/\sqrt{\log n}$ at fixed density. The stronger conditional proposals in the corpus remain conditional. A separate exact degree-flow lemma identifies a conserved algebraic resource, together with the representation and entry evaluation obligations still preventing its use for an order improvement. The report includes proofs, reproducible exact finite checks, immutable source provenance, and integration notes. The new results import the upstream deformation and recovery identities; this is not an independent certification of the claimed refutation of ETH.

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# 1 Outcome and the meaning of the improvement

The requested goal was to improve on the combined frontier of the recent proposals. The successful part of this investigation concerns the cost of evaluating their mode graph. Choosing its contraction structure after its exact index cardinalities are known gives a smaller certified exponential bound. The investigation did not establish a smaller logarithmic order. Both statements are essential to interpreting the result.

Put $\lambda = \lceil \log_2 s \rceil$ and $\mathcal{E}(s) = s/\sqrt{\lambda}$. The size $s$ counts normalized Boolean wires, not SAT variables. Let $\mathcal{L}$ denote the total input description length. The common interface consists of a bipartite Boolean tensor network with at most $s$ wires, maximum degree at most $\left\lceil s^{9/10} \right\rceil$, and entries in

$$(\mathbb{Z}/2^{P_0}\mathbb{Z})[V]/(V^{\ell}), \qquad 1 \le P_0, \ell \le s.$$

The original entry routines have time $(s+\mathcal{L})^{O(1)}2^{O(s^{9/10})}$ and polynomial space. The fixed constants in these requirements are part of the interface. The usual unramified extension of the scalar ring is used internally. Tensor descriptions retain their creation precisions and their original canonical representatives, as required by the source construction [1, 3].

**Theorem 1.1** (New relative contraction bounds)**.** *Under this interface and the imported exact identities listed in Section 3, there are deterministic exact algorithms with the following bounds:*

1. *Polynomial space and time* $(s+\mathcal{L})^{O(1)}2^{(C_{\mathrm{Q}}+o(1))\mathcal{E}(s)}$*, where*

$$50.30468921409 < C_{\mathrm{Q}} < 50.30468921410.$$

2. *Time* $(s+\mathcal{L})^{O(1)}2^{(C_{\mathrm{A}}+o(1))\mathcal{E}(s)}$ *and space* $(s+\mathcal{L})^{O(1)}2^{(C_{\mathrm{S}}+o(1))\mathcal{E}(s)}$*, where*

$$C_{\mathrm{A}} = C_{\mathrm{S}} = \frac{\pi}{\ln 2}\sqrt{40\log_2 9 \cdot \frac{2489356}{6638625}} = 31.2523973126509\ldots.$$

*The first algorithm is proved in Section 4; the second in Section 5. The* $o(1)$ *terms tend to zero for fixed interface constants. For example,* 50.305 *and* 31.253 *are safe eventual time coefficients, and* 31.253 *is also a safe eventual space coefficient for the second algorithm.*

| Common-interface certificate | Time coefficient | Space |
|---|---:|---:|
| Integrated round-two minimum | 76.90899495 | polynomial |
| Incoming streamed separators | 55.70947832 | polynomial |
| Stronger incoming composition | 55.40466591 | polynomial |
| New adaptive quartet algorithm | **50.30468922** | polynomial |
| Incoming cached contraction tree | 35.35458806 | $C_S = 34.27480301$ |
| New joint adaptive contraction | **31.25239732** | $C_S = 31.25239732$ |

Here a space coefficient means a bound $(s+\mathcal{L})^{O(1)}2^{(C_S+o(1))\mathcal{E}(s)}$. The two new algorithms represent different memory budgets; the cached one does not provide polynomial space. Lower-degree specializations in the corpus impose stronger input assumptions and are not comparable to the general-degree rows of this table.

The polynomial-space coefficient decreases by about 9.20% against the stronger composed baseline. The cached coefficient decreases by about 11.60% against the strongest displayed cached bound. More precisely, the ratios of the corresponding exponential certificate functions are

$$2^{-(5.0999766925+o(1))\mathcal{E}(s)}, \qquad 2^{-(4.1021907458+o(1))\mathcal{E}(s)}.$$

Both tend to zero. This comparison is between upper-bound functions. It proves neither a ratio of actual algorithm runtimes nor a practical speedup for SAT instances. The thresholds implicit in these constructions can be enormous.

### 1.1 Preserving the full existing envelope

Different packets also offer density-sensitive bounds, low-degree bounds, and other specialized interfaces. A finite algorithm portfolio can keep those methods and add the two new ones. On inputs satisfying several interfaces, interleaving the applicable exact algorithms incurs only a fixed scheduling overhead relative to the first to finish. A memory-restricted portfolio includes only algorithms respecting that restriction. Thus the new results can strengthen the common general-degree certificate while retaining every existing specialized option.

No such claim is made for an unimplemented conditional proposal. In particular, a hypothetical $n/\log n$ counter would asymptotically beat the present $n/\sqrt{\log n}$ scale. We have not constructed that counter.

## 2 Frozen source review

The initial repository snapshot is `fd0edd17e6a29117179b40a453c0def59165bd7c`, whose head commit is dated October 26, 2026 at 19:33:01 Pacific time. The review covers the requested `Computability/SubexponentialSAT` and `docs/incoming` directories at that snapshot. There are fifteen integrated refinement packets, five then-incoming archives, and two primary manuscripts. The machine-readable provenance files give exact Git blobs, SHA-256 hashes, paths, and immutable URLs. The local source checkout is unchanged.

A closing check reviewed commit `dde01c4a63b822c25b8eb345299346c36f837b49`, dated October 26 at 19:42:25 Pacific time. Its placement commit `450252e9d86c` files those same five archives under the Research directory. We checked all 230 archive members against the corresponding closing Git blobs; every member is byte-identical. The placement also restores two round-two replay scripts and updates the indexes. It adds no new mathematical packet. At this closing snapshot, all twenty packets are integrated and `docs/incoming` contains only its index. The initial snapshot ledger remains useful for identifying the exact arrival archives; `provenance/closing_review.json` records the verified new locations.

The source index explicitly labels this material as external research, with no ProveIt Lean or Rocq verification of these refinements [2]. We catalogued every packet's stated claim and inspected the proofs and interfaces used below. This does not mean that every proof in all twenty packets, or the full primary manuscript, has been independently verified.

### 2.1 Recent commits that determine the comparison

| Commit | Relevant change |
|---|---|
| `2b925d8800cd` | Places the first seven refinement packets and the separate Claude manuscript under SubexponentialSAT. |
| `3e096089911f` | Vendors the OpenAI preprint and its four LaTeX source files, pinned to `openai/math@7ef5c0cc5c`. |
| `45233238aebf` | Summarizes proved, density-sensitive, and conditional claims. |

| Commit | Relevant change |
|---|---|
| `6d3adc450615` | Places eight round-two packets; the smallest displayed general-degree coefficient is 76.90899. |
| `f42479bca245`, `f898a9fde4c3`, `fa6e904d109b` | Adds the five SAT archives that were incoming at the initial snapshot. Their stronger bounds must be included in the comparison. |
| `fd0edd17e6a2` | The initial pinned repository head used for the source extraction. |

## 2.2 The fifteen integrated packets

All paths in this subsection are relative to `Computability/SubexponentialSAT/Research/`. The exact report paths and fingerprints are in `provenance/source_inventory.json`.

| Packet | Claim relevant to the frontier |
|---|---|
| `cauchy_mode_improvement` | Factorial Cauchy modes with asymptotic weight slope three; coefficient improvement, unchanged order. |
| `eth_contraction_improvements` | Sharp Hermite taper, precision $6MD/u$, streamed coefficient extraction, schedule coefficient about 648.44. |
| `3sat_precision_refinement` | Exact precision and taper refinements. The $n \log\log n/\log n$ and $n/\log n$ targets require an unproved structured-summation algorithm. |
| `sat_density_improvement` | Uniform density envelope with numerator $n \min\{\log(2+m/n), \log\log n\}$. |
| `eth_density_refinement` | Overlapping density improvement and a local slope-four Cauchy Smith profile. |
| `3sat_refinement` | Scalar parity lift and smaller incidence degree; faster second contraction remains conditional on a compact factorization. |
| `3sat_asymptotic_analysis` | Restricted enumeration/taper barriers; degree-preserving and linear-wire interfaces remain missing. |
| `eth_asymmetric_packet_balancing` | Asymmetric side-degree packing; general-degree $C = 76.90899$, stronger-assumption low-degree $C = 31.06413$. |
| `3sat_round2_refinement` | Two-stage schedule, $C = 84.06627$; also actual-domain and alternate-manuscript audits. |
| `eth_balanced_packing` | Nearly lossless padding and additive precision guard; $C = 94.74820$. |
| `eth_high_moment` | Higher-moment balancing; the same $C = 94.74820$, with low-degree specializations. |
| `eth_matching_refinement` | Matching-potential balancing; again $C = 94.74820$. |
| `eth_packet_refinement` | Two-stage labeling, thin precision gaps; $C = 95.66394$. |
| `eth_recursive_refinement` | Residue-domain modes and an improved schedule; $C = 133.20737$. |
| `eth_joint_cauchy` | Whole-block slope-four integral separation, exact local Smith spectrum, and cached mode-graph contraction; $C = 319.68576$ in polynomial space or 225.59706 with subexponential space. |

The table records the reports' claims, including their explicit conditions. Several already combine into stronger coefficients than their displayed standalone figures. We use those combinations when defining our baseline.

## 2.3 The five incoming archives

Each archive name below has suffix `_2026-10-26.zip`. All five were safely extracted and reviewed. Their original archives and every extracted member are identified in the provenance ledger.

| Archive stem | Strongest pertinent statements |
|---|---|
| `sat_adaptive_contraction` | Adaptive triples: cached time 36.75031252, space 34.02417801. Streamed binary separators: polynomial-space time 55.70947832. Also a finite modular-kernel implementation. |
| `sat_frontier_improvements` | Variable-arity schedules and cache composition. Polynomial-space coefficient about 63.8574, cached coefficient 38.48449591. |
| `subexponential_sat_frontier` | Heterogeneous rational schedules: polynomial-space 63.85220339, cached 37.93010908; stronger comparisons against composed baselines. |
| `subexponential_sat_kernel_improvements` | Adaptive Smith-mode selection and Gray, antipodal, and projective coefficient-extraction improvements; local guarantees, with unchanged global order. |
| `subsat_contraction_tree` | Recursively built block tables and a coordinate/CRT bridge to a 12/5 matrix exponent: cached time 35.35458806, space 34.27480301. |

All five retain the original logarithmic exponent order. The polynomial-space separator algorithm of the first archive can be combined with the heterogeneous schedule of the third. This yields the stronger baseline 55.40466590661, rather than merely 55.70947831891. Our exact checker recomputes it. Within our own new schedule, replacing adaptive quartet conditioning by fixed binary conditioning gives 54.02470271284; this matched comparison isolates the gain from the new graph evaluator.

The cached result also requires a fair comparison. It improves on the smallest displayed incoming coefficient 35.35458806 while decreasing its space coefficient. It is a new certified point, not a proof of global optimality over every conceivable retuning or composition of prior ideas.

### 2.4 SAT-level bounds and conditional targets

The two primary manuscripts [1, 12] state the general-density scale

$$(n+m)^{O_k(1)}2^{O_k(n\log\log n/\sqrt{\log n})}.$$

The OpenAI Appendix D also states the fixed-density scale $2^{O_{k,D}(n/\sqrt{\log n})}$. The density packets [11] refine the common envelope to

$$(n+m)^{O_k(1)}2^{O_k\left(\frac{n}{\sqrt{\log n}}\min\{\log(2+m/n),\log\log n\}\right)}. \tag{1}$$

For example, density $m/n=\Theta(\log\log n)$ replaces a $\log\log n$ numerator by $\log\log\log n$ in the corresponding density bound. Our methods can be used within this framework, but the displayed order in (1) is unchanged.

The stronger proposal $O(n\log\log n/\log n)$, with fixed-density $O(n/\log n)$, is conditional on an unproved summation interface [10]. Other proposals need an unconstructed degree-preserving or linear-wire compiler. Their hypothetical conclusions should neither be advertised as implemented algorithms nor claimed to be surpassed here. The separate Claude manuscript does not provide a matched explicit wire coefficient. We make no numerical dominance claim over an unknown constant in that different architecture.

# 3 The exact mode-graph interface

## 3.1 Imported algebra and precision

We import the upstream Cauchy deformation, integrality and Hermite recovery, the child-network factorization, and the immutable creation-precision convention [1]. We import the refined Hermite leading loss, the asymmetric near-uniform packing conditions, and the whole-block slope-four integral modes [3, 4, 5]. These imported claims are stated as hypotheses of the relative refinement. The new graph arguments do not change their exact algebraic identities.

At an active call let $M$ be its wire count, $u$ its retained taper range, $L = 2^a$ its packet size, and $D = 2^{a-i}$ at retained valuation level $i < u$. For $r$ groups, the imported estimates are

$$p_t \le (6 + o(1))MD/u, \tag{2}$$

$$c_e \le \left(\frac{2}{r^2} + o(1)\right) M/D \quad \text{for each directed group pair } e, \tag{3}$$

$$w_q = a + 1 - i + 4q - 2s_2(q). \tag{4}$$

The number $c_e$ bounds the number of modes of index $q$ on pair $e$; $s_2(q)$ is the binary popcount. Every mode has one source and one target. Child networks have $(2/r + o(1))M$ wires. All $o(1)$ errors used below are uniform over active depths under the stated schedules.

Fix an array of exact pair weights $b_e \ge 0$ with $\sum_e b_e < p_t$. For fixed $r$ there are polynomially many arrays. Let $\mathcal{I}_e$ be the list of nonnegative mode-exponent vectors of exact weight $b_e$ on pair $e$. Skip any array with an empty list. The exact contribution is

$$Z = \sum_{(\xi_e) \in \prod_e \mathcal{I}_e} \prod_{g=1}^{r} Z_g((\xi_e)_{e \ni g}). \tag{5}$$

An evaluation of $Z_g$ is a genuine recursive tensor contraction. It is charged in all recurrences below, rather than treated as a constant-time oracle. Exact list cardinalities, enumeration and rank/unrank are available from the integer dynamic program

$$A(j, b) = A(j - 1, b) + A(j, b - w_j), \qquad A(0, 0) = 1,$$

with other zero or negative boundary cases assigned zero. Mode counts, budgets and count bit lengths are polynomial in $s$.

## 3.2 A uniform Euclidean entropy bound

Set $h_e = \log_2 |\mathcal{I}_e|$, and

$$\mathcal{K}(M, u) = \kappa M/\sqrt{u}, \qquad \kappa = \pi\sqrt{2}/\ln 2.$$

The adaptive algorithms require an estimate uniform in the actual weight array, not just a bound for a single predetermined set of edges.

**Lemma 3.1** (Uniform entropy norm)**.** *For each fixed group count $r$ in our schedules,*

$$\|h\|_2 \le (1/r + o(1))\mathcal{K}(M, u). \tag{6}$$

*Consequently any fixed finite family of possibly data-selected nonnegative vectors $z$ satisfies $\sum_e z_e h_e \le (\|z\|_2/r + o(1))\mathcal{K}(M, u)$.*

*Proof.* Lower the weights to $1+4q-2s_2(q)$ and extend their range to all $q \geq 0$. Their counting function is $x/4 + O(\log(x+2))$, and therefore

$$\sum_{q\geq 0} -\ln(1-e^{-\theta(1+4q-2s_2(q))}) = \frac{\pi^2}{24\theta} + O(\log^2(2+1/\theta)), \quad 0 < \theta \leq 1.$$

This follows by integrating against the counting function and the identity $\int_0^\infty -\ln(1-e^{-x})\,dx = \pi^2/6$. If $c$ is the common upper envelope in (3), the generating product gives

$$\ln|\mathcal{I}_e| \leq \theta b_e + c\frac{\pi^2}{24\theta} + O(c\log^2(2+1/\theta)).$$

For $b_e > 0$ take $\theta = \min\{1, \sqrt{c\pi^2/(24b_e)}\}$. When the minimum equals one, the excess cost is $O(c)$. The case $b_e = 0$ has a singleton list. As all budgets are polynomial, this proves the uniform estimate

$$h_e \leq \frac{\pi}{\sqrt{6}\ln 2}\sqrt{cb_e} + O(c\lambda^2).$$

Taking the Euclidean norm and using $\sum b_e < p_t$ yields

$$\|h\|_2 \leq \frac{\pi}{\sqrt{6}\ln 2}\sqrt{cp_t} + O_r(c\lambda^2) \leq (1/r + o(1))\mathcal{K}(M,u).$$

Indeed every schedule uses the additive buffer $b = \lceil 6\log_2\lambda\rceil$ and $a_d = \sum_{j\leq d} u_j + (d+1)b$, with $a_{-1} = 0$, so $D \geq 2^{a_{d-1}+b+1} \geq \lambda^6$. The last error divided by $M/\sqrt{u}$ is $O(\lambda^{5/2}/D) = o(1)$. This also handles zero and arbitrarily small pair budgets. Cauchy–Schwarz proves the final assertion without choosing an unjustified asymptotic saddle for each individual list. □

## 4 Adaptive quartet conditioning in polynomial space

The incoming adaptive-contraction packet [5] already uses balanced binary separator conditioning. The new step in this section is to choose the binary pairing of four fixed blocks from the actual exact-weight list sizes. This choice is repeated inside the blocks. It changes the worst-case coefficient of the paid child-call count; it is not merely a retuning of the taper schedule.

We use the common normalized contraction interface and its imported asymmetric packing, slope-four Cauchy separation, precision, and child construction results. For one exact pair-budget array, let $\mathcal{I}_{xy}$ be the list on directed group pair $(x,y)$, discard an array with an empty list, and put

$$h_{xy} = \log_2|\mathcal{I}_{xy}|, \qquad \mathcal{K}(M,u) = \kappa\frac{M}{\sqrt{u}}, \qquad \kappa = \frac{\pi\sqrt{2}}{\ln 2}.$$

Lemma 3.1 implies, for every fixed nonnegative coefficient vector $z$ on the directed group pairs,

$$\sum_{x\neq y} z_{xy}h_{xy} \leq \left(\frac{\|z\|_2}{r} + o(1)\right)\mathcal{K}(M,u). \tag{7}$$

All coefficient vectors used below belong to a finite family independent of $s$, since every alphabet size is fixed. Thus the bound is uniform even though our algorithm selects a vector indirectly from the list sizes. This avoids an unjustified uniformity inference for an arbitrary continuously varying $z$.

### 4.1 The four-block identity and its norm bound

Partition a set of groups into four fixed nonempty blocks $B_1, B_2, B_3, B_4$. Let

$$H_{ab} = \sum_{x \in B_a,\, y \in B_b} (h_{xy} + h_{yx}) \quad (a < b)$$

be the logarithm of the product of list sizes crossing a pair of blocks. There are three perfect matchings of the four blocks. They partition the six unordered block-pair classes into three complementary pairs. In the $j$th such pair write its two weights as $a_j \geq b_j$.

For a pairing $(B_a \cup B_b) \mid (B_c \cup B_d)$, condition first on all directed indices between the two sides, then condition inside each side between its two constituent blocks. The largest external index weight on a path ending at one of the four block oracles is the total cross-block weight minus $\min(H_{ab}, H_{cd})$. Consequently the best pairing has exact external path bound

$$F_4(H) = \sum_{j=1}^{3} (a_j + b_j) - \max_j b_j. \tag{8}$$

Its choice is computable without real logarithms: for each matching take the smaller of its two integer products of list sizes, and maximize that integer. There are just three candidates.

**Lemma 4.1** (Adaptive quartet inequality)**.** *For every nonnegative six-edge weighting $H$,*

$$F_4(H) \leq \sum_j a_j + \frac{2}{3} \sum_j b_j \leq \sqrt{\frac{13}{3}} \left( \sum_{a<b} H_{ab}^2 \right)^{1/2}.$$

*Both inequalities can hold with equality. For four singleton groups, there is a directed-edge coefficient vector of squared norm $26/3$ whose scalar product with $h$ bounds the maximum path weight of the selected separator tree.*

*Proof.* Since $\max_j b_j \geq \frac{1}{3} \sum_j b_j$, the first inequality follows from (8). Cauchy–Schwarz gives the second because $3 \cdot 1^2 + 3 \cdot (2/3)^2 = 13/3$. Equality occurs when the three $a_j$ are all $3t$ and the three $b_j$ are all $2t$, for any $t \geq 0$. Each undirected singleton-pair weight is the sum of two directed weights; the coefficient pattern therefore has squared norm $2(13/3) = 26/3$. □

The lemma is sharp for this local family of three balanced binary trees. It does not claim that every possible contraction algorithm has this worst-case coefficient.

For unequal block sizes $k_a = |B_a|$, a weighted version is useful. Choose any fixed probabilities $p_1, p_2, p_3 \geq 0$ with sum one. Equation (8) implies

$$F_4(H) \leq \sum_j a_j + \sum_j (1 - p_j) b_j.$$

The complementary block-pair classes in matching $j$ contain respectively $c_j$ and $d_j$ directed pairs, each of the form $2k_a k_b$. Regardless of which class has the larger actual weight, the squared coefficient norm is at most

$$R(k, p) = \sum_{j=1}^{3} \left[ \max(c_j, d_j) + \min(c_j, d_j)(1 - p_j)^2 \right]. \tag{9}$$

The probabilities are only an analysis device; the algorithm still chooses the matching by its exact score in (8).

### 4.2 Recursive quartets and all child calls

Apply the same rule inside each of the four blocks. Blocks of size two use their unique binary split; singletons are leaves. Let $g_r$ bound the squared norm of one coefficient vector that upper-bounds the largest root-to-leaf index weight of the resulting tree on $r$ groups. If the four block sizes are $k_1, \dots, k_4$, then

$$g_r \le R(k,p) + \max_a g_{k_a}, \qquad g_1 = 0, \quad g_2 = 2. \tag{10}$$

Indeed, the external maximum is bounded by the quartet vector. The largest internal path cost occurs in one block; choose its inductively constructed vector. Internal and external supports are disjoint, so squared norms add. The total cost is bounded by the sum of the two maxima, even if they occur at different leaves. This possible looseness is harmless for an upper bound.

For four equal blocks of size $k$, use $p_j = 1/3$. Then

$$g_{4k} \le \frac{26}{3}k^2 + g_k.$$

For $r = 4^j$ this yields

$$g_r = \frac{26}{45}(r^2 - 1), \tag{11}$$

where equality denotes our certificate, not an optimality assertion. The fixed balanced binary-tree certificate is $2(r^2-1)/3$; its ratio to (11) is exactly $15/13$. Thus the new square-root coefficient is smaller by $\sqrt{13/15}$ for these alphabets.

The schedule below needs only the following exact ledger:

| $r$ | four block sizes | $(p_1, p_2, p_3)$ | $g_r$ | fixed binary $v(r)$ |
|---|---|---|---|---|
| 4 | $(1,1,1,1)$ | $(1/3, 1/3, 1/3)$ | 26/3 | 10 |
| 6 | $(1,1,2,2)$ | $(0, 1/2, 1/2)$ | 22 | 24 |
| 8 | $(2,2,2,2)$ | $(1/3, 1/3, 1/3)$ | 110/3 | 42 |
| 16 | $(4,4,4,4)$ | $(1/3, 1/3, 1/3)$ | 442/3 | 170 |

For $r = 6$, the complementary directed cardinalities are $(2,8)$, $(4,4)$, and $(4,4)$. Their contribution in (9) is 20, and the largest internal block contributes $g_2 = 2$.

At an internal node of a separator tree, enumerate the directed indices between its two children, recursively evaluate each conditional sum, multiply the two answers, and accumulate. At a singleton, evaluate the group factor by its genuine recursive tensor contraction. Finite distributivity proves the value is exactly the original mode sum over any commutative coefficient ring. If $E_v$ is the union of cuts on the path to group $v$, its factor is called precisely $\prod_{e \in E_v} |\mathcal{I}_e|$ times. Hence every child call is paid, and (7) gives the local recurrence

$$T_d \le \mathrm{poly}(s) 2^{(\sqrt{g_{r_d}}/r_d + o(1))\mathcal{K}(M_d, u_d)} T_{d+1}. \tag{12}$$

All loops, ranks, exact counts, current assignments, and ring accumulators have polynomial descriptions. Depth-first traversal uses polynomial space. There are no cached exponential tables and no free group-factor oracle in this bound.

### 4.3 A strict rational schedule

Let $\lambda = \lceil \log_2 s \rceil$. Use

| $d$ | 0 | 1 | 2 | 3 | $4 + k$ $(k \ge 0)$ |
|---|---|---|---|---|---|
| $r_d$ | 16 | 16 | 8 | 6 | 4 |
| $\eta_d$ | 1249/100000 | 1249/100000 | 26/3125 | 107/20000 | $(27/6400)(5/8)^k$. |

(13)

Put $u_d = \lfloor \eta_d \lambda \rfloor$, use the additive buffer $b = \lceil 6 \log_2 \lambda \rceil$, set $a_d = \sum_{j \le d}(u_j + b)$ with $a_{-1} = 0$, and stop the network recursion at $M \le H = \lceil s/\lambda \rceil$. The cumulative leading ranges are

$$A_0 = \frac{1249}{100000}, \quad A_1 = \frac{1249}{50000}, \quad A_2 = \frac{333}{10000}, \quad A_3 = \frac{773}{20000}, \quad A_\infty = \frac{499}{10000}.$$

At the root the symmetric packing gate is $2A_0 \log_2 r_0 < 1/10$; below it the old-side asymmetric gate is $A_d \log_2 r_d < 1/10$. The exact integer inequality $6^{200} < 2^{517}$ gives $\log_2 6 < 517/200$. Using this bound for six, the margins at the five alphabet regimes are at least

$$\frac{1}{12500}, \quad \frac{1}{12500}, \quad \frac{1}{10000}, \quad \frac{359}{4000000}, \quad \frac{1}{5000},$$

respectively. The other packing exponents are bounded by $2/10 + A_\infty = 2499/10000 < 1$. Newly created degrees have exponent at most $A_\infty + o(1) < 9/10$. These are strict fixed margins.

Here is a noncircular depth justification. Verify the gates first for hypothetical active depths through $\lceil \log_{5/3} \lambda \rceil + O(1)$. Throughout that range,

$$u_d \ge \lambda^{\epsilon + o(1)}, \qquad \epsilon = 1 - \frac{\ln(8/5)}{\ln(5/3)} > 0,$$

since $8/5 < 5/3$. Thus floors are negligible, and the total buffers cost only $O((\log \lambda)^2) = o(\lambda)$. The strict gates imply nearly lossless padding, and eventually every child has at most $3M/5$ wires. Consequently no active path extends beyond the proposed range. The sharper shrink ratios give, uniformly on active paths,

$$M_d \le (1 + o(1)) s f_d, \quad (f_0, f_1, f_2, f_3) = \left(1, \frac{1}{8}, \frac{1}{64}, \frac{1}{256}\right), \quad f_{4+k} = \frac{1}{768}\, 2^{-k}.$$

The inherited precision bound is $O(s\lambda^2 2^{a_{d-1}})$. The buffer gives $D \ge 2^{a_{d-1}+b+1}$, so its ratio to $MD/u$ is $O(\lambda^{-2})$ while $M > H$. This establishes the sharp precision and entropy premises uniformly, including at the final active depths.

**Theorem 4.2** (Polynomial-space adaptive-quartet certificate)**.** *Under the normalized source contraction interface, adaptive quartet conditioning with* (13) *takes*

$$(s + \mathcal{L})^{O(1)}\, 2^{(C_{\text{quartet}} + o(1)) s / \sqrt{\lceil \log_2 s \rceil}}$$

*time and polynomial space, where*

$$C_{\text{quartet}} = \kappa \Bigg[ \frac{9\sqrt{442/3}}{128\sqrt{1249/100000}} + \frac{\sqrt{110/3}}{512\sqrt{26/3125}} + \frac{\sqrt{22}}{1536\sqrt{107/20000}} + \frac{\sqrt{26/3}}{3072\sqrt{27/6400}\,(1 - \sqrt{2/5})} \Bigg],$$

*with the certified interval*

$$50.30468921409 < C_{\text{quartet}} < 50.30468921410.$$

*In particular,* 50.305 *is a valid eventual coefficient.*

*Proof.* Sum the logarithms of the branching factors in (12) along a path. The tail has ratio $(1/2)/\sqrt{5/8} = \sqrt{2/5}$; substituting the ledger gives the displayed expression. The $O(\log \lambda)$ levels of polynomial factors contribute $O(\lambda \log \lambda)$ to the logarithm of time. Leaves cost $2^{O(s/\lambda)}$, initial entry routines cost $2^{O(s^{9/10})}$, and freshly created packet routines cost $\exp(s^{499/10000 + o(1)})$. Each is lower order than $s/\sqrt{\lambda}$, including after paying ancestor child-call factors. Canonical creation-time descriptions prevent repeated expansion of old entry routines. Depth-first traversal proves the space claim. The coefficient interval follows by exact rational arithmetic in the accompanying checker. □

For a particularly simple structural comparison, keep the incoming adaptive packet's fixed $r = 16$ schedule unchanged. Replacing its balanced binary trees by adaptive quartets gives

$$C_{16} = \frac{5\pi}{2\ln 2}\sqrt{\frac{221}{15}}\left(1 + \frac{1}{3\sqrt{3}}\right) = 51.862701863893\ldots,$$

exactly $\sqrt{13/15}$ times its previous $55.70947831891\ldots$ coefficient. The improved strict schedule then lowers this to 50.30468921410.

### 4.4 Comparison and verification boundary

The two compatible incoming methods already compose to a stronger polynomial-space comparator than either headline alone: combine the heterogeneous $(13, 14, 8, 6, 4, \ldots)$ schedule in the frontier packet with the adaptive packet's fixed binary separators. This gives $55.40466590661\ldots$. A numerical search suggests a fixed-tree $(16, 16, 8, 6, 4, \ldots)$ boundary of $54.00208\ldots$, but it is not needed for any proof or global optimality claim. At the same strict rational schedule (13), the old fixed trees have coefficient $54.02470271284\ldots$, whereas the new quartet trees have $50.30468921410\ldots$. This matched comparison isolates the new structural gain from schedule retuning.

The checker performs 15,625 exhaustive four-group inequalities, 2,000 exact integer-weight profile checks on $r = 4, 6, 8, 16$, and 80 literal-versus-streamed contractions over $(\mathbb{Z}/64\mathbb{Z})[V]/(V^2)$. It also verifies every individual leaf-call count (14,056 calls in total), the unequal-block norm ledgers, strict capacity inequalities, and rational coefficient bounds. Three matched fixtures additionally reduce leaf-factor calls from 64 to 16, 128 to 32, and 256 to 64 for four, eight, and sixteen groups, respectively, with exactly equal outputs; these are operation counts, not timings. All tests remain active under optimized Python. These finite checks support the implementation and numerical certificates; the all-size statement rests on the proofs above and the explicitly imported source interfaces. There is no new proof-assistant build or complete SAT engine. The result improves a normalized-wire exponent coefficient and preserves the polynomial-space guarantee; the SAT logarithmic order is unchanged.

## 5 A joint covariance bound for cached contraction

The strongest cached incoming certificate has time coefficient $35.3545880584\ldots$ and space coefficient $34.2748030050\ldots$ on $s/\sqrt{\log_2 s}$. Its binary table construction and fast triangle contraction are both useful, but their separate worst-case entropy bounds need not occur on the same group partition. The new algorithm chooses one partition by the largest of *all* its construction and triangle costs. A covariance calculation controls this joint objective and pays for every intermediate table and every recursively evaluated child.

We use the exact pair-index identity (5) and Lemma 3.1; write $K(M, u) = \mathcal{K}(M, u)$ in this section. Every group function is evaluated by a paid recursive child contraction. The uniform bound needed for partition selection is

$$\|h\|_2 \le (1/r + o(1))K(M, u). \tag{14}$$

The source of the cached interface and exact quotient-ring matrix bridge is the incoming contraction-tree packet [7].

### 5.1 Six costs that must be controlled together

Let $r = 3k$ and equipartition the groups into $A, B, C$, each of size $k$. Write $I_A, I_B, I_C$ for the total directed-pair entropy incident to each block, counting an internal pair once. Write $H_{AB}, H_{AC}, H_{BC}$ for

the three crossing entropies, and put $S = H_{AB} + H_{AC} + H_{BC}$. Define

$$\mathcal{C}(\mathcal{P}) = (I_A, I_B, I_C, S - \tfrac{3}{5}H_{AB}, S - \tfrac{3}{5}H_{AC}, S - \tfrac{3}{5}H_{BC}), \quad W(\mathcal{P}) = \max_{1\le j\le 6} \mathcal{C}_j(\mathcal{P}). \tag{15}$$

The first three costs permit direct construction of the three block tables from cached group functions. The largest of the last three is $S - (3/5)\min(H_{AB}, H_{AC}, H_{BC})$, the rectangular matrix multiplication cost supplied by an exponent-12/5 backend. The integer-coordinate/CRT bridge in the incoming report [7] makes that backend valid over the actual power-of-two quotient ring. Here are its bit-cost and algebraic requirements. The ring is a free module of polynomial rank over $\mathbb{Z}/2^p\mathbb{Z}$, with a polynomial-size integer multiplication table. Lift coordinates to integers, compute each coordinate-pair matrix product modulo sufficiently many nonexceptional odd primes, reconstruct by CRT, apply the structure constants, and reduce modulo $2^p$. An unreduced entry has bit length $O(p + \log d)$ for inner dimension $d$, which is polynomial in $s$. A fixed rational bilinear algorithm of exponent below 12/5 exists by the established field bound [8]; excluding its finitely many denominator primes makes its identities valid over these fields. Coordinate arithmetic and CRT add only polynomial bit overhead. Sequential recursion uses a polynomial factor times the input/output table sizes in storage. The algorithm never inverts two, a nilpotent, or an arbitrary nonunit in the source ring.

**Lemma 5.1** (Joint adaptive partition bound)**.** *For $r = 9$ or $r = 12$, an equipartition can be selected so that*

$$W(\mathcal{P}) \le r\beta_r \|h\|_2, \qquad \beta_9^2 = \frac{2489356}{6638625}, \qquad \beta_{12}^2 = \frac{4786024}{13540725}.$$

*In particular $\beta_9 = 0.6123566464\ldots$ and $\beta_{12} = 0.5945200108\ldots$.*

*Proof.* Combine opposite directed weights into undirected weights $x_{\{g,h\}} = h_{gh} + h_{hg}$. Put $H = \sum_e h_e = \sum_e x_e$, $N = r(r-1)$, and $m = N/2$. Choose a uniformly random labelled equipartition. For one undirected edge its vector of coefficients in (15) is, up to permuting block labels,

$$v_{\text{internal,A}} = (1, 0, 0, 0, 0, 0), \qquad v_{AB} = (1, 1, 0, 2/5, 1, 1).$$

Center each such six-vector by subtracting its coordinate mean; call the result $\widetilde{v}_e$. If $\overline{\mathcal{C}} = \frac{1}{6}\sum_j \mathcal{C}_j$ and $V = \sum_j (\mathcal{C}_j - \overline{\mathcal{C}})^2$, then

$$\mathbb{E}\overline{\mathcal{C}} = A_r H, \quad A_r = \frac{1 + (17/5)p_r}{6}, \quad p_r = \frac{2k}{r-1}, \qquad \mathbb{E}V = \sum_{e,f} x_e x_f \, \mathbb{E}\langle \widetilde{v}_e, \widetilde{v}_f \rangle.$$

The covariance entry depends only on whether $e = f$, whether the edges share one endpoint, or whether they are disjoint. Denote the three entries by $c_0, c_1, c_2$. They are the following exact rational numbers:

| $r$ | $c_0$ | $c_1$ | $c_2$ | $A_r$ |
|---|---|---|---|---|
| 9 | 109/120 | 869/4200 | −379/2100 | 71/120 |
| 12 | 299/330 | 181/750 | −3151/24750 | 191/330 |

For completeness, these entries require only a fixed finite color calculation. Take edges $(1,2)$ and respectively $(1,2)$, $(1,3)$, $(3,4)$. For their $v = 2, 3, 4$ distinct endpoints, sum over the $3^v$ colorings $\chi$ with colors $A, B, C$. If $n_j$ endpoints receive color $j$, the probability is

$$\frac{(k)_{n_A}(k)_{n_B}(k)_{n_C}}{(3k)_v}, \qquad (a)_t = a(a-1)\cdots(a-t+1).$$

Multiply this by the dot product of the two centered coefficient vectors and sum. This evaluates the table using at most 81 terms per entry. For example $c_0 = (1 - p_r)(5/6) + p_r(14/15)$. The accompanying checker evaluates every term as a rational number.

Let $x'_e = x_e - H/m$ and $d'_v = \sum_{e \ni v} x'_e$. The covariance matrix has constant row sum

$$L_0 = c_0 + 2(r-2)c_1 + \tfrac{1}{2}(r-2)(r-3)c_2.$$

On the centered weights its quadratic form is $(c_0 - 2c_1 + c_2)\|x'\|_2^2 + (c_1 - c_2)\|d'\|_2^2$. Cauchy–Schwarz gives $\|d'\|_2^2 \le (r-2)\|x'\|_2^2$: use $\sum_v (d'_v)^2 = \sum_{\{v,w\}} x'_{vw}(d'_v + d'_w)$ and $\sum_v d'_v = 0$. Since $c_1 - c_2 \ge 0$,

$$\mathbb{E}V \le L_1 \left( \|x\|_2^2 - \frac{H^2}{m} \right) + L_0 \frac{H^2}{m}, \quad L_1 = c_0 + (r-4)c_1 - (r-3)c_2.$$

The values are

$$(L_0, L_1) = \begin{cases} (3/200, 1059/350), & r = 9, \\ (1/275, 5476/1375), & r = 12. \end{cases}$$

In both cases $0 \le L_0 < L_1$. Because $\|x\|_2^2 \le 2\|h\|_2^2$, it follows that

$$\mathbb{E}V \le 2L_1\|h\|_2^2 - \frac{2(L_1 - L_0)}{N} H^2.$$

For six numbers, the largest is at most their mean plus $\sqrt{5/6}$ times their centered Euclidean norm. Jensen therefore implies, with $q = 1 - L_0/L_1 > 0$,

$$\mathbb{E}W \le A_r H + \sqrt{5L_1/3}\sqrt{\|h\|_2^2 - qH^2/N} \le \sqrt{\frac{A_r^2 N}{q} + \frac{5}{3}L_1}\,\|h\|_2.$$

The last step is Cauchy–Schwarz with coordinates $\sqrt{q}\,H/\sqrt{N}$ and $\sqrt{\|h\|_2^2 - qH^2/N}$. A minimum-cost partition is no worse than the mean. Substitution of the rational table gives $\beta_r^2 = r^{-2}(A_r^2 N/q + 5L_1/3)$, and the displayed fractions. □

**Exact partition selection.** All objectives can be compared without logarithms. For each block let $D_I$ be the product of the exact list sizes incident to it, and let $D_{AB}, D_{AC}, D_{BC}$ be the three crossing dimensions. Minimize the maximum of the six rational scores

$$D_{I_A}^5, \quad D_{I_B}^5, \quad D_{I_C}^5,$$
$$\frac{(D_{AB}D_{AC}D_{BC})^5}{D_{AB}^3}, \quad \frac{(D_{AB}D_{AC}D_{BC})^5}{D_{AC}^3}, \quad \frac{(D_{AB}D_{AC}D_{BC})^5}{D_{BC}^3}.$$

These scores have logarithms $5\mathcal{C}_j$. There are only 280 unlabelled equipartitions for $r = 9$, and 5775 for $r = 12$. Their counts and their exact integer bit lengths are polynomially bounded in $s$.

### 5.2 Every cache, table, and recursive call is paid

First cache all group functions $Z_g$ on their incident indices. The number of child calls has exponent at most $(\alpha_r + o(1))K(M,u)$, where $\alpha_r = \sqrt{2(r-1)}/r$, by (14). Each child has at most $(2/r + o(1))M$ wires.

After choosing the partition, form a block table by summing the product of its cached group functions over its internal pair indices. Literal enumeration costs at most a polynomial factor times $2^{I_B}$. A sequential implementation expands each factor into records on all incident indices of the block, sorts on the common assignment key, multiplies the matched records, then sorts and sums on the boundary key. Each expanded list has at most $2^{I_B}$ records. Thus the claimed bit work and space do not rely on

free random access to exponential tables. Boundary indices are a subset of incident indices, so the output table also has at most $2^{I_B}$ entries.

Contract the three tables by the imported rectangular matrix backend. Its time exponent is the largest of the three triangle costs in (15); its input/output dimensions and its sequential workspace are bounded by the same exponent. Every group cache is no larger than the incident-index set of the block containing that group. The selected $W$ consequently bounds *all* local storage as well as all nonrecursive work. Equations (14) and Lemma 5.1 prove the local recurrence

$$T_d \le \mathrm{poly}(s)\left(2^{(\alpha_{r_d}+o(1))K_d}T_{d+1} + 2^{(\beta_{r_d}+o(1))K_d}\right), \qquad S_{\mathrm{local}} \le \mathrm{poly}(s)2^{(\beta_{r_d}+o(1))K_d}. \tag{16}$$

For the small late alphabets $r = 6, 4$ below, a simpler cached algorithm suffices: cache all group functions and sum over all pair indices. It has the same child-call coefficient $\alpha_r$, local time coefficient $\beta_r^{\mathrm{full}} = \sqrt{r(r-1)}/r$, and local storage coefficient at most $\alpha_r$.

### 5.3 A multistage schedule with strict packing margins

Put $\lambda = \lceil \log_2 s \rceil$, $g = \lceil \sqrt{\lambda} \rceil$, and $b = \lceil 6 \log_2 \lambda \rceil$. Let $B$ be the largest nonnegative integer with $9^{20B} \le 2^{\lambda - g}$, using direct contraction for bounded exceptional sizes. Define

| $d$ | $r_d$ | $u_d$ | upper cumulative retained range |
|---|---|---|---|
| 0 | 9 | $B$ | $(\lambda - g)/(20 \log_2 9)$ |
| 1 | 12 | $\lfloor \lambda/36 \rfloor - B$ | $\lambda/36$ |
| 2 | 9 | $\lfloor \lambda/288 \rfloor$ | $\lambda/32$ |
| 3 | 6 | $\lfloor 5\lambda/864 \rfloor$ | $\lambda/27$ |
| $d \ge 4$ | 4 | $\lfloor \lambda/(189 \cdot 2^{d-4}) \rfloor$ | $\lambda/21$ |

Set $a_d = \sum_{j \le d}(u_j + b)$ with $a_{-1} = 0$, and leaf threshold $H = \lceil s/\lambda^{3/4} \rceil$. The cumulative caps use $1/36 + 1/288 = 1/32$, $1/32 + 5/864 = 1/27$, and $1/27 + 2/189 = 1/21$.

At the root the symmetric term $Q_0^2\Delta$ has binary logarithm at most $\lambda - g/10 + O(\log \lambda)$. At later depths the retained-side condition has exponent $9/10 + (\log_2 r_d)a_d/\lambda$. Strict rational margins follow from

$$12^{100} < 2^{359}, \quad 9^{40} < 2^{127}, \quad 6^5 < 2^{13},$$

which give respectively margins $1/3600$, $1/1280$, and $1/270$ at depths $1, 2, 3$; the constant-4 tail has margin $1/210$. The fresh-side and padding exponents are at most $(2 \log_2 r_d + 1)a_d/\lambda < 1$ throughout. Fresh packet degrees have exponent at most $1/21 + o(1) < 9/10$.

For a noncircular bootstrap, consider hypothetical active depths through $4 + \frac{3}{4}\log_{9/5} \lambda + O(1)$. The tail ranges satisfy

$$u_d \ge \lambda^{\eta + o(1)}, \qquad \eta = 1 - \frac{3 \ln 2}{4 \ln(9/5)} > 0,$$

since $2^3 < (9/5)^4$. All packing gates hold uniformly on this range. Nearly lossless padding then gives eventual tail shrinkage at most $5/9$, closing the hypothesized depth bound. The sharp masses on a path are, up to uniformly vanishing multiplicative perturbations,

$$m_0 = 1, \quad m_1 = 2/9, \quad m_2 = 1/27, \quad m_3 = 2/243, \quad m_{4+t} = \frac{2}{729\, 2^t}, \qquad M_d \le (1 + o(1))^d s m_d.$$

The additive buffer gives $D \ge 2^{a_{d-1}+b+1}$ at retained levels. The inherited precision estimate $O(s\lambda^2 2^{a_{d-1}})$ is $o(MD/u)$ because $s/M \le \lambda^{3/4}$ and $\lambda^{15/4}2^{-b} \to 0$. Thus the entropy bounds used above are uniform at every active call.

### 5.4 Certified total time and space

Let $\kappa = \pi\sqrt{2}/\ln 2$ and $A = 1/(20\log_2 9)$. On the scale $s/\sqrt{\lambda}$, the $K_d$ coefficients are

$$K_0^* = \kappa/\sqrt{A}, \quad K_1^* = \frac{2\kappa}{9\sqrt{1/36 - A}}, \quad K_2^* = \frac{\kappa\sqrt{288}}{27}, \quad K_3^* = \frac{2\kappa}{243}\sqrt{864/5}, \quad K_{4+t}^* = \frac{2\kappa\sqrt{189}}{729}\,2^{-t/2}.$$

Write $P_d = \sum_{j<d} \alpha_{r_j} K_j^*$ for $0 \le d \le 4$. Expanding (16) gives the following complete list of competing coefficients:

| source of work | coefficient |
|---|---|
| root local work | $\beta_9 K_0^* = 31.25239731265\ldots$ |
| depth 1 local work | $P_1 + \beta_{12} K_1^* = 30.41173930826\ldots$ |
| depth 2 local work | $P_2 + \beta_9 K_2^* = 30.23123727305\ldots$ |
| depth 3 full pair sum | $P_3 + (\sqrt{30}/6) K_3^* = 30.18781782003\ldots$ |
| entire constant-4 tail | $P_4 + K_4^* \max\{\sqrt{12}/4, (\sqrt{6}/4)/(1 - 1/\sqrt{2})\} = 30.42570652863\ldots$ |

Exact rational interval arithmetic in the checker verifies every strict comparison. Hence

$$C_{\text{joint}} = \frac{\pi\sqrt{2}}{\ln 2}\sqrt{20\log_2 9}\sqrt{\frac{2489356}{6638625}} = 31.2523973126509\ldots \tag{17}$$

bounds the total time coefficient. Local storage is bounded by $\beta_9 K_0^*$, $\beta_{12} K_1^*$, $\beta_9 K_2^*$, and the smaller late singleton caches; ancestor tables add rather than multiply. The same $C_{\text{joint}}$ is therefore a valid space coefficient. This space bound is an upper certificate, not an assertion that the largest table attains it.

Polynomial factors per level add only $O(\lambda \log \lambda)$ to the time exponent. Direct leaves cost $2^{O(s/\lambda^{3/4})}$, original entry routines $2^{O(s^{9/10})}$, and fresh packet routines at most $\exp(s^{1/21+o(1)})$, all lower order even after the paid recursive call counts. Immutable creation-time descriptions prevent hidden composition of entry routines. We obtain, under the stated inherited network interface,

$$T(s, \mathcal{L}), S(s, \mathcal{L}) \le \text{poly}(s + \mathcal{L}) 2^{(C_{\text{joint}} + o(1))s/\sqrt{\log_2 s}}.$$

The eventual rounded coefficient 31.253 is certified. The method uses subexponential space. It decreases a leading exponential coefficient, while leaving the worst-case and fixed-density SAT logarithmic orders unchanged.

## 6 Conserved degree and recursive representation

The incidence-degree route has a useful invariant stronger than the statement that every recursive object has degree at most a fixed global bound. The invariant is algebraic; it does not by itself give the local entry algorithms required by the contraction theorem.

### 6.1 Degree transfer between groups

Let $R$ be a commutative ring and let $G$ be a finite directed multigraph on groups $1, \ldots, r$, with no loops. Each group has a disjoint set $X_g$ of commuting indeterminates. For an edge $e$, write $s(e)$ and $t(e)$ for its source and target, and introduce a mode indeterminate $Z_e$. Choose

$$P_g \in R[X_g, (Z_e)_{t(e)=g}], \qquad \deg P_g \le K_g, \qquad L_e \in R[X_{s(e)}], \quad \deg L_e \le 1.$$

Here degree counts both local variables and mode variables, and $K = \sum_g K_g$. For a nonnegative integer tuple $\beta = (\beta_e)$ put

$$b_g^{\mathrm{in}} = \sum_{t(e)=g} \beta_e, \qquad b_g^{\mathrm{out}} = \sum_{s(e)=g} \beta_e,$$

and define

$$Q_{g,\beta}(X_g) = \left[ \prod_{t(e)=g} Z_e^{\beta_e} \right] P_g \prod_{s(e)=g} L_e(X_g)^{\beta_e}.$$

**Degree-flow lemma.** One has the finite polynomial identity

$$\prod_g P_g(X_g, (L_e(X_{s(e)}))_{t(e)=g}) = \sum_\beta \prod_g Q_{g,\beta}(X_g). \tag{18}$$

For every nonzero sector contribution,

$$\deg Q_{g,\beta} \le d_{g,\beta} := K_g - b_g^{\mathrm{in}} + b_g^{\mathrm{out}}, \qquad \sum_g d_{g,\beta} = K. \tag{19}$$

The caps are nonnegative whenever all relevant coefficients are nonzero. Replacing every $Q_{g,\beta}$ by its multilinear representative on a Boolean cube preserves its Boolean values and does not increase its degree.

**Proof.** Expand each $P_g$ as a polynomial in its incoming $Z$ variables. Each mode occurs as an incoming variable at exactly one target; substituting its linear form contributes its power at exactly one source. Regrouping these factors gives (18). Extracting an incoming coefficient of total mode degree $b_g^{\mathrm{in}}$ leaves local degree at most $K_g - b_g^{\mathrm{in}}$; multiplying the outgoing linear forms adds at most $b_g^{\mathrm{out}}$. Finally every $\beta_e$ is subtracted once and added once, proving the sum identity. Boolean multilinearization replaces a positive power by its first power and can only lower degree. These arguments remain valid over rings with zero divisors, including the truncated coefficient rings in the source; degree bounds can only decrease through cancellations. □

The groupwise Boolean sums therefore obey

$$\sum_{x \in \{0,1\}^{\bigsqcup_g X_g}} \prod_g P_g(x_g, (L_e(x_{s(e)}))_{t(e)=g}) = \sum_\beta \prod_g \left( \sum_{x_g \in \{0,1\}^{X_g}} Q_{g,\beta}(x_g) \right).$$

Thus incoming modes remove exactly the degree allowance that outgoing mode powers add elsewhere. It is unnecessary to charge the full original degree separately to every child. If a global degree budget is presented through a grading rather than groupwise bounds, the statement applies after fixing a feasible group-degree allocation; for fixed $r$, there are at most $(K+1)^r$ such allocations. This observation does not make the resulting local polynomials easy to evaluate.

### 6.2 Why the low-degree deformation is valid

For a multilinear polynomial $A$ in $M$ variables, the elementary multilinear interpolation identity is

$$\sum_{Y \in \{0,1\}^M} A(Y) \prod_{x=1}^{M} l_{Y_x}(z_x) = A(z), \qquad l_0(z) = 1 - z, \quad l_1(z) = z.$$

Consequently, if $\deg A \le K$ and the entries of the source deformation matrix $W(T)$ have degree at most $2L-1$, its deformed contraction can be written

$$F(T) = \sum_{X \in \{0,1\}^M} B(X) A(W(T)X), \qquad \deg_T F \le K(2L-1).$$

The tensor $B$ need not have low polynomial degree for this conclusion: its Boolean entries do not depend on $T$. In an original-variable representation, COPY constraints can be enforced by keeping a single Boolean variable wherever it occurs. In such a representation the degree-flow lemma describes the formal effect of a mode sector.

### 6.3 The precise closure gap

There are three distinct representations, with different obligations. First, the original incidence polynomial has a controlled degree, but may have $N = n(\log n)^{O(1)}$ incidence positions. Its direct Cauchy mode multiplicity is controlled by $N/D$, rather than by the compressed bit count divided by $D$. Second, the binary subset-index representation has $O(n \log\log n)$ wires, but incidence degree does not bound the Boolean degree of its index tensors. Third, the formal group polynomial $Q_{g,\beta}$ has the degree bound (19), but its coefficient extraction can couple all factors within a group.

The source's ordinary child network retains the original row and column vertices together on one side and puts new packet tensors on the other. This preserves packetwise entry evaluation. It does not preserve the assertion that a single designated side has total low polynomial degree. Alternatively, carrying $Q_{g,\beta}$ as one implicit local factor preserves the degree statement but has not supplied the required entry algorithm. Expanding the entire group polynomial can take exponential time in its group degree. This is a cost of that direct method, not an unconditional lower bound on all possible implementations.

The following elementary identity illustrates the distinction. For any $d$ by $d$ matrix $A = (a_{ij})$ over a commutative ring,

$$[Z_1 \cdots Z_d] \prod_{i=1}^{d} \left( \sum_{j=1}^{d} a_{ij} Z_j \right) = \sum_{\pi \in S_d} \prod_{i=1}^{d} a_{i,\pi(i)} = \operatorname{perm}(A). \tag{20}$$

All input factors are linear and have short descriptions, but grouping them creates a global coefficient-extraction task. Equation (20) is an algebraic illustration only: it is not an assertion that every matrix occurs as an actual Cauchy mode block, nor a complexity lower bound for those blocks.

An adequate recursive compiler must simultaneously preserve the degree budget, the compressed mode multiplicity, a suitable graph-degree bound, polynomial-size immutable descriptions, and subdominant entry evaluation at the inherited creation precisions. The degree-flow lemma supplies one of these invariants and identifies the exact incoming/outgoing transfer. It does not supply the compiler, and no improved worst-case SAT exponent is deduced here.

**Relation to the inspected corpus.** The existing `3sat_asymptotic_analysis` packet [9] already states a conditional degree-preserving kernel interface and explains why binary subset indices do not inherit incidence degree. The present note isolates an explicit conserved group-degree identity and its coefficient-extraction obligation. A text search of the integrated and five incoming packets found no statement of this identity in this form. This is a statement about the inspected corpus, not a claim of mathematical priority; the identity follows from elementary polynomial expansion.

# 7 Why the SAT exponent order is still unchanged

The new algorithms accelerate a finite graphical sum that was already present in the contraction proof. Their local costs remain $2^{\Theta(M/\sqrt{u})}$ at the level of the proved upper-bound expressions. The admissible taper range remains proportional to $\log s$, because packet label counts and the high-degree packing gates must remain polynomially bounded. The complete recurrence therefore still yields a certificate of the form $2^{(C+o(1))\mathcal{E}(s)}$. This observation describes the present analysis, not a lower bound on the true running time of these algorithms or on every possible algorithm.

The two distinct routes to a better order should be kept separate. One is to evaluate the mode graph faster than its present entropy scale, using structure beyond arbitrary pair-index factors. The other is to decrease the effective size or deformation degree before the graph is formed, while preserving a usable recursive representation. The degree-flow identity in Section 6 contributes to the second route. It supplies an exact conserved quantity but not the required compiler.

Taking an infimum of constants, changing the base of a logarithm, using a stronger matrix multiplication exponent, or omitting a lower-order entry routine does not by itself establish $o(s/\sqrt{\log s})$ in the exponent. Similarly, an unproved summation oracle cannot be inserted into a recurrence and then counted as a completed improvement. We retain the explicit resource accounting throughout the new proofs.

# 8 Reproducibility and verification

The archive contains the complete LaTeX source, compiled PDF, original Python prototypes, exact numerical certificates, source fingerprints, an independent review, and rebuild instructions. The scripts use Python's standard library. The exploratory numerical schedule searches are not dependencies of any theorem or certificate.

## 8.1 What the executable checks establish

The quartet checker supplies a generic exact finite pair-graph evaluator, an exhaustive reference evaluator, and an instrumented child-call ledger. Its checks include 15,625 four-group inequalities, 2,000 weighted tree profiles, and 80 exact contraction comparisons over the nonfield ring $(\mathbb{Z}/64\mathbb{Z})[V]/(V^2)$. Every one of the 14,056 leaf calls in those comparisons agrees with the predicted product of index-domain sizes. Three additional fixtures reduce calls $64 \to 16$, $128 \to 32$, and $256 \to 64$, with equal nonzero outputs. These fixtures validate exact work counts; they are not wall-clock measurements or SAT benchmarks.

The joint-objective checker verifies all endpoint-color covariance sums for nine and twelve groups, 28 integer-weight objective examples, and 27 exact finite mode-graph contractions modulo 2048. Its rational interval computation certifies every competing recurrence path and each packing gate. Independently written code enumerates all 1,680 labelled partitions for nine groups and all 34,650 for twelve groups, confirming the same covariance constants and testing additional integer-weight examples. A separate cached-block prototype checks 24 contractions over $(\mathbb{Z}/64\mathbb{Z})[V]/(V^2)$, including all 945 group-oracle calls and 1,052 block-construction visits in its fixtures. The prototypes use literal final summation to check equality; they do not implement or benchmark the theoretical fast-matrix backend.

The degree-flow checker verifies 80 exact polynomial examples, 2,598 mode sectors, 640 Boolean evaluations, 48 permanent coefficient identities, and 40 low-degree deformation identities. A negative control explicitly detects the incorrect rule that omits outgoing-mode degree. This tests the algebraic invariant without pretending to supply the missing recursive entry evaluator.

The numerical certificates use rational upper and lower bounds for $\pi$ and logarithms and integer-square-root brackets. Ordinary floating point is not needed to establish the coefficient inequalities. All checks use explicit exceptions rather than removable Python assertions.

### 8.2 Proof and implementation status

The new separator and covariance inequalities are proved for all nonnegative weights. The finite tests support the corresponding implementations and arithmetic ledgers; they do not prove the asymptotic theorems by sampling. The global contraction consequences retain the imported identities and hypotheses in Section 3. No new Lean or Rocq formalization was built, and no complete subexponential SAT solver was implemented or benchmarked.

The source manifest fingerprints 347 files in the pinned requested folders and the vendored preprint, plus all 230 members of the five incoming ZIPs. Fingerprinting establishes source identity; it is not proof verification. The comparison ledger identifies all twenty packets and their stronger-assumption or conditional statements.

## 9 Further research questions

**1. The optimal adaptive separator constant.** For a fixed group count $r$, define

$$\Gamma_r = \sup_{h\geq 0,\, \|h\|_2=1} \min_{T} \max_{v} \sum_{e\in E_v(T)} h_e,$$

where $T$ ranges over binary separator trees and $E_v(T)$ is the union of its cut-pair sets along the path to $v$. The quartet inequality is sharp for four singleton groups within its three balanced trees, but the repeated-quartet bound need not be sharp for larger $r$. A proof of a smaller $\Gamma_r$, with a constructive selector and all expensive leaf calls paid, would improve the polynomial-space coefficient again. Such a result would still need a separate argument to change the logarithmic order.

**2. Choose table construction and final contraction together.** The six-cost theorem shows why forming block tables and contracting them should be optimized together. A larger finite family of full computation trees may yield a better covariance or minimax bound. It could include binary block construction, direct construction, rectangular-matrix choices, and memory constraints under one objective. Exact pair cardinalities make the finite search computable; a new analysis must control all stages under the same selected partition.

**3. Compile the conserved degree budget.** The degree-flow identity permits group degree to move along a mode without increasing its global sum. Can a representation preserve that budget, compressed mode multiplicity, bounded graph degree, and subdominant entry evaluation simultaneously? A useful positive result must construct the entry evaluator. A useful negative result must be restricted to a precisely specified representation or oracle model. The permanent identity alone is not such a lower bound.

**4. Exploit actual sparse crossing blocks.** The incoming adaptive Smith-mode packet reduces the mode inventory for some actual real-port matrices. It cannot increase the full padded kernel's Smith slope uniformly. It remains possible that the specific blocks generated by a SAT reduction have additional sparsity or rank structure that combines with the present adaptive graph evaluation.

The question should be studied on recursively generated blocks, with preprocessing and valuation-normalization costs included.

**5. Close the verification gap at the right interface.** A compact formal development can begin with the mode-graph identity, exact list indexing, quartet selection, and finite covariance formulas. These are smaller and more isolated obligations than an end-to-end rebuild of the upstream SAT claim. A subsequent implementation should measure actual mode-graph workloads and keep those results distinct from the theoretical fast-matrix backend and from SAT-level timings.

The completed deliverable is a stronger pair of explicit contraction certificates and an exact algebraic invariant for the next order-improvement attempt. The stronger global asymptotic target remains open in this work.

## References

[1] OpenAI. *The Exponential Time Hypothesis is False*. October 25, 2026. Preprint pinned at `openai/math@7ef5c0cc5c`; Appendix D and `build/quantitative-contraction.tex`. https://github.com/openai/math/tree/7ef5c0cc5ca4b0e072b7a050fca09ab24bc6fa3a/preprints/The-Exponential-Time-Hypothesis-is-False-October-25-2026. Vendored source fingerprints are included in this archive.

[2] Vladimir Reshetnikov. ProveIt SubexponentialSAT project index and research collection, pinned commit `fd0edd17e6a2`. https://github.com/VladimirReshetnikov/ProveIt/tree/fd0edd17e6a29117179b40a453c0def59165bd7c/Computability/SubexponentialSAT.

[3] *Asymmetric packet balancing for subexponential SAT*. Research packet, October 26, 2026. https://github.com/VladimirReshetnikov/ProveIt/blob/fd0edd17e6a29117179b40a453c0def59165bd7c/Computability/SubexponentialSAT/Research/eth_asymmetric_packet_balancing/report.tex.

[4] *Full Crossing-Block Cauchy Modes*. Research packet, October 26, 2026. Full-block integral Cauchy separation and slope-four mode profile. https://github.com/VladimirReshetnikov/ProveIt/blob/fd0edd17e6a29117179b40a453c0def59165bd7c/Computability/SubexponentialSAT/Research/eth_joint_cauchy/report.tex.

[5] *Adaptive aggregation for subexponential SAT*. Incoming archive `sat_adaptive_contraction_2026-10-26.zip`, member `adaptive_sat_review.tex`. Streamed separators, adaptive triples, exact pair budgets and precision accounting. https://github.com/VladimirReshetnikov/ProveIt/blob/fd0edd17e6a29117179b40a453c0def59165bd7c/docs/incoming/sat_adaptive_contraction_2026-10-26.zip.

[6] *Beyond the Combined SAT Proposal Frontier*. Incoming archive `subexponential_sat_frontier_2026-10-26.zip`, especially `theory.tex`. Heterogeneous schedules and stronger composed comparisons. https://github.com/VladimirReshetnikov/ProveIt/blob/fd0edd17e6a29117179b40a453c0def59165bd7c/docs/incoming/subexponential_sat_frontier_2026-10-26.zip.

[7] *Contraction trees and streamed Cauchy features for subexponential SAT*. Incoming archive `subsat_contraction_tree_2026-10-26.zip`, especially `report/cached_tree.tex`. Exact pair indices, recursive table construction, and the integer-coordinate/CRT matrix bridge. https://github.com/VladimirReshetnikov/ProveIt/blob/fd0edd17e6a29117179b40a453c0def59165bd7c/docs/incoming/subsat_contraction_tree_2026-10-26.zip.

[8] Josh Alman, Ran Duan, Virginia Vassilevska Williams, Yinzhan Xu, Zixuan Xu, and Renfei Zhou. *More Asymmetry Yields Faster Matrix Multiplication*. arXiv:2404.16349v2, October 20, 2024; SODA 2025. The version-specific bound $\omega < 2.371339$ suffices for the strict inequality $\omega < 12/5$ used here. https://arxiv.org/html/2404.16349v2.

[9] *Asymptotic improvement analysis for the October 2026 3-SAT preprint*. Integrated packet `3sat_asymptotic_analysis`, particularly the degree-kernel interface and representation barriers. https://github.com/VladimirReshetnikov/ProveIt/tree/fd0edd17e6a29117179b40a453c0def59165bd7c/Computability/SubexponentialSAT/Research/3sat_asymptotic_analysis.

[10] *Sharper precision for subexponential 3SAT*. Integrated packet `3sat_precision_refinement`. The stronger summation rate is explicitly conditional. https://github.com/VladimirReshetnikov/ProveIt/tree/fd0edd17e6a29117179b40a453c0def59165bd7c/Computability/SubexponentialSAT/Research/3sat_precision_refinement.

[11] *Improving the Clause-Density Dependence of Exact k-SAT Counting* and *A density-sensitive refinement of the October 2026 exact #k-SAT counting algorithm*. Integrated packets `sat_density_improvement` and `eth_density_refinement`, October 26, 2026. Immutable report URLs and file hashes for both packets, and every other packet in the review, appear in `provenance/source_inventory.json`.

[12] *A $2^{o(n)}$ time algorithm for 3-SAT*, credited to Claude, October 26, 2026. The separate 49-page manuscript `Research/claude-2o-n-3sat/eth.pdf` in the pinned collection. Its implementation and unspecified constants are not independently certified by this report.